

BIS 629 Lecture 1: Theory of hypothesis testing

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1. Specification of the hypothesis:

a. The standard hypothesis

The hypothesis test: $H_0 : \beta^* = 0$ vs. $H_A : \beta^* \neq 0$

- β^* is the parameter of interest, also called the effect estimate, the treatment effect, the intervention effect
- This hypothesis test is a two-sided test, because under the alternative, we take an agnostic view and allow for β^* to be either positive or negative.

The two-sided test generally has less power than the one-sided test,
 $H_0 : \beta^* < 0$ vs. $H_A : \beta^* > 0$

- In implementation science, and biomedical research in general, one-sided tests are rarely used.

1. Specification of the hypothesis:

a. The standard hypothesis

Why two-sided tests?

There have been surprises, where a treatment believed to be beneficial turned out to be harmful

There are many examples, but here is one:

The Beta-Carotene and Retinol Efficacy Trial (CARET) tested the effect of daily vitamins A and C on the incidence of lung cancer in 18,314 high risk participants. CARET was stopped ahead of schedule because participants who were randomly assigned to the intervention had a 28% increase in lung cancer incidence, a 17% increase in death and higher cardiovascular disease mortality compared with the placebo group.

Reference: Effects of a combination of beta carotene and vitamin A on lung cancer and cardiovascular disease. Omenn GS, Goodman GE, Thornquist MD, Balmes J, Cullen MR, Glass A, Keogh JP, Meyskens FL, Valanis B, Williams JH, Barnhart S, Hammar S. N Engl J Med. 1996 May 2;334(18):1150-5.

Unless otherwise noted, in this course, we will be considering two-sided tests.

Example

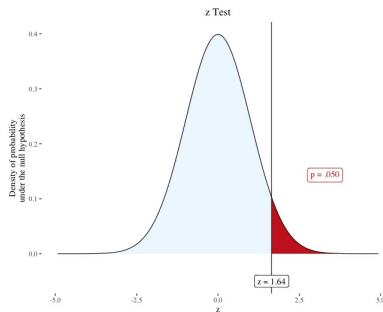
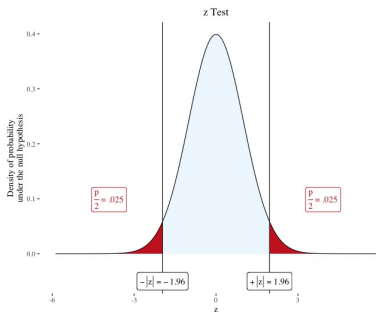
We compare the difference between sample means in an intervention and control group. Here, β^* , the effect estimate, is $\mu_2 - \mu_1$, where μ_1 is the mean of the outcome in the control group and μ_2 is the mean of the outcome in the intervention group.

Under the two-sided test with Type I error rate = 5%, we reject H_0 when the test statistic, here the Z-score,

$$Z = \frac{\hat{\mu}_2 - \hat{\mu}_1}{\sqrt{\widehat{Var}(\hat{\mu}_2 - \hat{\mu}_1)}}$$

is > 1.96 or < -1.96 , the 2.5th and 97.5th percentiles of the standard normal distribution, $N(0, 1)$.

This means, that, when the null hypothesis is true, it will be rejected incorrectly 5% of the time. Type I error rates of 5% are nearly universal in implementation science and other biomedical research.



Under the one-sided test, we reject H_0 when the Z-score is > 1.645 , the 95th percentile of the standard normal distribution, $N(0, 1)$.

Example: Familia Salama

Goal: Increase the number of antenatal care (ANC) visits during pregnancy to the World Health Organization (WHO) recommended standard of 4 or more during the pregnancy

β^* is the difference between the number of ANC visits among women in the intervention wards compared to the number of ANC visits among women in the control wards –

- estimated with continuous, non-normally distributed outcome data

$$H_0 : \beta^* = 0 \text{ vs. } H_A : \beta^* \neq 0$$

Example: Familia Salama

Let $Y_{ij} = 1$ if woman i in ward j had 4 or more ANC visits, 0 otherwise.

Now β^* is the difference in the proportion of participants with ≥ 4 ANC visits among women in the intervention wards vs. the control wards – $H_0 : \beta^* = 0$ vs.

$H_A : \beta^* \neq 0$

- estimated with binary data, identity link function – difference between two binomial proportions

Example: Familia Salama

Or β^* is the ratio:

β^* is the ratio of the proportions ≥ 4 ANC visits among women in the intervention wards vs. the control wards –

- estimated with binary data, log or logistic link function – ratio or odds ratio between two binomial proportions:

$$H_0 : \beta^* = 1 \text{ vs. } H_A : \beta^* \neq 1$$

$\equiv$

$$H_0 : \frac{p_2}{p_1} = 1 \text{ vs. } H_A : \frac{p_2}{p_1} \neq 1$$

That is, β^* is defined as $\frac{p_2}{p_1}$.

Note that by simple re-arrangement,

$$H_0 : \frac{p_2}{p_1} = 1 \equiv H_0 : p_2 = p_1 \equiv H_0 : p_2 - p_1 = 0$$

1. Specification of the hypothesis:

b. The non-inferiority hypothesis

In implementation science, we may also be interested in the non-inferiority hypothesis. In the related discipline of comparative effectiveness research, the non-inferiority hypothesis is of considerable importance.

For example, HLB-Simple trial, Uganda, HIV+ patients under HIV care – 3 arms

- 1 Standard of care for hypertension control
- 2 HTN-Basic: receive free and consistent access to diagnostic equipment and evidence-based antihypertensive drugs
- 3 HTN-Plus: HTN-Basic plus a multi-component implementation strategy to improve uptake and adherence to BP treatment, contextually adapted to these clinics, using a human-centered design approach to adapt differentiated service delivery models, use of hypertensive PLHIV peer champions, and methods of blood pressure monitoring to address specific barriers to, and facilitators of, blood pressure control.

- The primary effectiveness outcome will be % of patients with hypertension (HTN) diagnosis who are controlled ($< 140\text{mmHg}$ systolic and $\leq 90\text{ mmHG}$ diastolic)— estimated as a binomial proportion
- A question of interest is whether HTN-Plus improves hypertension control compared to HTN-BASIC
- Now, this question can be formulated as a standard null hypothesis as above, where β^* = the difference in proportions under HTN control in each arm, and the null hypothesis is that HTN-BASIC, p_1 , equals the proportion under control in HTN-PLUS, p_2 .
- Alternatively, we can investigate whether HTN-BASIC is not inferior to HTN-PLUS, vs. the alternative that it is inferior, with non-inferiority specified by a parameter δ . For example, it might be reasonable to test the hypothesis that

$H_0 : p_1 - p_2 \geq \delta$ (new intervention is inferior, where δ could be 0) vs.
 $H_A : p_1 - p_2 < \delta$ (new intervention is not inferior)

Then, $Z = \frac{(\hat{p}_1 - \hat{p}_2 - \delta)}{\sqrt{\hat{Var}(\hat{p}_1) + \hat{Var}(\hat{p}_2)}}$, and $p - \text{value} = 1 - \Phi(Z)$, where $\Phi(\cdot)$ is the cumulative standard normal

Non-inferiority and equivalence tests

- Perhaps HTN-PLUS is a little better, but it may not be worth the time effort, expense unless it improves hypertension control compared to HTN-BASIC by, for example, $\delta = 5$ percentage points or more.
- Another way to write a non-inferiority hypothesis uses one-sided confidence intervals.
- Here, we write $H_0 : p_1 - p_2 + 1.645\sqrt{\text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2)} > \delta$ inferior vs.
 $H_0 : p_1 - p_2 + 1.645\sqrt{\text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2)} \leq \delta$ not inferior
- In clinical research, non-inferiority tests are of interest as well
- Here the concern is that a new intervention might improve a primary outcome that is beneficial on a primary outcome, but may also have adverse side effects.
- Then, we may wish to test the null hypothesis that the new intervention is 'worse' than the control with respect to this adverse outcome, vs. the alternative that it is not worse (non-inferior)

Non-inferiority and equivalence tests

- For example, in Familia Salama, Option B may be more effective in preventing HIV transmission from mother to child, but there is concern it may also cause birth defects.
- Let p_1 be the rate of birth defects among the babies of mothers who received option A, and p_2 be the same for Option B. We would not like the birth defect rate to be more than $\delta = 0.5\%$ higher than Option A in the Option B arm. Thus, $\delta = 1.005 \times p_1$
- Hence, $H_0 : p_2 - p_1 \geq \delta$ vs. $H_A : p_2 - p_1 < \delta$
- Alternatively, one could specify the null hypothesis such that the upper bound of the 95% confidence interval for $p_2 - p_1$ is $\geq \delta$, that is,
$$H_0 : p_2 - p_1 + 1.645\sqrt{\text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2)} \geq \delta \text{ vs.}$$
$$H_A : p_2 - p_1 + 1.645\sqrt{\text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2)} < \delta$$

Equivalence tests

- Equivalence tests are one-sided tests against a known standard. For example, if we knew that the birth defect rate in the general population was $\delta = 0.01\%$, we might wish to assess the consistency of the data from Option B with population birth defect rates
- That is, $H_0 : p_2 \geq \delta$ vs. $p_2 < \delta$
- Alternatively, one could specify the null hypothesis such that the upper bound of the 95% confidence interval for p_2 is $\geq \delta$, that is,
 $H_0 : p_2 + 1.645\sqrt{\text{Var}(\hat{p}_2)} \geq \delta$ vs. $H_A : p_2 + 1.645\sqrt{\text{Var}(\hat{p}_2)} < \delta$

Any hypothesis can be specified. Then it remains to derive, given the study design and available data, how to construct the test statistic and assess its significance.

Construction of the test statistic

- We need to assess the consistency of the data generated in the study with the null hypothesis.
 - if it's not consistent with the null hypothesis we reject it and conclude the alternative hypothesis is more consistent with the data, that is, the test was 'significant'.
- To do this we construct a test statistic.
 - The test statistic is a function of the data with a known **asymptotic** distribution under the null hypothesis and under the alternative.
- We will consider asymptotic tests only in this course because implementation science tends to work with large studies. Situations where asymptotic results may be less reliable will be discussed.

The test statistic, $Z^2(\mathbf{Y}, \mathbf{X}; \beta)$ is a function of the data, $(\mathbf{Y}, \mathbf{X})$, where

- $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)^T$, the outcomes for participants $j = 1, \dots, n$ in the study, with n the total sample size, and
- $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$ is the intervention/treatment status for the study participants.
- β is a vector of parameters, where β^* is the parameter of interest in β . The number of parameters is q .
- Further notation will be introduced as needed as the course progresses.

Classic test statistics

The two classic test statistics are

- the test for the difference between two sample means, and
- the test for the difference between two binomial proportions
- Many other test statistics are derivatives of these two

The **test for the difference between two sample means** has the following form

$$Z^2 = \frac{(\hat{\mu}_2 - \hat{\mu}_1)^2}{\widehat{Var}(\hat{\mu}_2 - \hat{\mu}_1)},$$

where

$\hat{\mu}_1 = \sum_{j=1}^{n_1} Y_j / n_1$, and $Y_j, j = 1, \dots, n_1$ are the continuous outcome data for Arm 1 of the study, and similarly for $\hat{\mu}_2, j = n_1 + 1, \dots, n_1 + n_2$, where $n_1 + n_2 = n$

Test for the difference between two sample means

- Asymptotically, that is, as $n_1 \rightarrow \infty$ and $n_2 \rightarrow \infty$, under the central limit theorem, under H_0 ,

$$Z^2 \xrightarrow{d} \chi_1^2,$$

a central χ^2 distribution with 1 degree of freedom.

And under H_A ,

$$Z^2 \xrightarrow{d} \chi_1^2(\lambda),$$

a non-central central χ^2 distribution with 1 degree of freedom and non-centrality parameter, λ , where $\lambda = \frac{(\beta_{H_A}^*)^2}{\text{Var}(\hat{\mu}_2 - \hat{\mu}_1)}$

- These asymptotic results apply **regardless** of the distribution of $\mathbf{Y} = (\mathbf{Y}_1^T, \mathbf{Y}_2^T)$, where $\mathbf{Y}_1 = (Y_1, \dots, Y_{n_1})^T$ and $\mathbf{Y}_2 = (Y_{n_1+1}, \dots, Y_{n_1+n_2})^T$.
- These results will also apply if $\mathbf{Y}_1 \sim N(\mu_1, \sigma_1^2)$ and $\mathbf{Y}_2 \sim N(\mu_2, \sigma_2^2)$ exactly, that is, if the outcome data from both groups are normally distributed.
- We will provide theoretical justification for the distributional assumptions for this test statistic using both likelihood arguments and methods of moments/generalized estimating equations arguments soon

Example: Familia Salama, Dar es Salaam, Tanzania

Group	Intervention	SoC	Total
Sample size	N=134,938	N=108,980	N=243,918
Proportion of ≥ 4 ANC visits	12,560/134,938 (9.3%)	7,155/108,980 (6.6%)	19,715/243,918 (8.1%)
ICC (95% CI) for Proportion of ≥ 4 ANC visits	0.05 (0.03, 0.07)	0.07 (0.05, 0.11)	0.05 (0.03, 0.07)
Mean (SD) for number of ANC visits	1.79 (1.21)	1.67 (1.06)	1.74 (1.15)
ICC (95% CI) for number of ANC visits	0.07 (0.05, 0.10)	0.10 (0.07, 0.15)	0.06 (0.04, 0.08)

example1.png

$$Z^2 = \frac{(1.79 - 1.67)^2}{1.21^2/134938 + 1.06^2/108980} = 680.52$$

$$p\text{-value} = 1 - Pr(Z^2 \leq \chi^2_{1,\alpha} | H_0) = 0$$

Why is this p-value wrong? We will correct this later.

Asymptotic theory for test statistics

- Let β is the parameter vector indexing the function of the data upon which the test statistic is based, where usually in implementation science, the parameter of interest, β^* , is a scalar element from this vector, with the remaining elements being nuisance parameters.
- For a given scenario, the challenge is to find a test statistic that has a known probability distribution under the null and its alternative.
- When considering asymptotic test statistics, we need to identify a test statistic that converges in law to a standard $N(0, 1)$ under H_0 . Then, asymptotically, the square of this test statistic follows the central and non-central χ^2 distribution under H_0 and H_A respectively, providing two-sided critical values in a straightforward manner.

3 standard approaches

- Wald test
- Likelihood ratio test
- Score test

Let the log-likelihood function be

$$\sum_{j=1}^n \log f(Y_j | X_j; \beta),$$

where $f(Y_j | X_j; \beta)$, the probability density model that is assumed to have generated the data, (Y_j, X_j) , has a form such that it includes β^* , the parameter of interest, as β^* .

Likelihood-based approaches for constructing test statistics

- The maximum likelihood estimator, MLE, $\hat{\beta}$, is

$$\arg \max_{\beta} \sum_{j=1}^n \log f(Y_j | X_j; \beta),$$

- which is equivalent to the solution with respect to β of

$$\frac{\partial \sum_{j=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta} = \mathbf{0}$$

Likelihood-based approaches for constructing test statistics

- The asymptotic variance of $\hat{\beta}^*$, is the r - r element of the variance-covariance matrix

$$\text{Var}(\hat{\beta}) = \left[-E(\mathbf{Y}|\mathbf{X}) \left\{ \frac{\partial^2 \sum_{j=1}^n \log f(Y_j|X_j; \beta)}{\partial \beta \partial \beta^T} \right\} \right]^{-1}.$$

Note that this variance is not a function of $\mathbf{Y}$ and β^* is the r th element of β .

- Sometimes people write **AVar** instead of **Var** to indicate the asymptotic variance. Because we only consider asymptotic variances in this class, we will write **Var**.

1. Wald test

$$Z^2 = \frac{(\hat{\beta}^*)^2}{\text{Var}(\hat{\beta}^*)}$$

where $\text{Var}(\hat{\beta}^*)$ was defined above.

This test is asymptotically equivalent to

$$Z^2 = \frac{(\hat{\beta}^*)^2}{\widehat{\text{Var}}(\hat{\beta}^*) \Big|_{\beta^* = \hat{\beta}^*}}.$$

- *Asymptotically equivalent* means that the statistics converge in probability to the same value and have the same asymptotic distribution with the same variance. They are not the same functions of the data, that is, they are not equal.

1. Wald test

- Under H_0 , $Z^2 \xrightarrow{d} \chi_1^2$.
- Under H_A , $Z^2 \xrightarrow{d} \chi_1^2(\lambda)$, where λ , the non-centrality parameter, is
$$\lambda = \frac{(\beta_{HA}^*)^2}{\text{Var}(\hat{\beta}^*)}$$
- Example: Logistic regression
 - Here, $f(Y_j|X_j) = \frac{\exp(\beta_0 + \beta^* X_j)^{Y_j}}{1 + \exp(\beta_0 + \beta^* X_j)}$ and the expected value of Y_j given X_j is $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$,
 - The log-likelihood for the study with data $\{\mathbf{Y}, \mathbf{X}\}$ of sample size $n = n_1 + n_2$,

$$\begin{aligned} L(\beta) &= \sum_{j=1}^n \log f(Y_j|X_j; \beta) \\ &= \sum_{j=1}^n Y_j(\beta_0 + \beta^* X_j) - \log\{1 + \exp(\beta_0 + \beta^* X_j)\} \end{aligned} \quad (1)$$

1. Wald test

- The asymptotic variance of $\hat{\beta}^*$, is the 2 – 2 element of the variance-covariance matrix

$$\text{Var}(\hat{\beta}) = \left[-E(\mathbf{y}|\mathbf{x}) \left\{ \frac{\partial^2 \sum_{j=1}^n \log f(Y_j|X_j; \beta)}{\partial \beta \partial \beta^T} \right\} \right]^{-1}$$

1. Wald test

From eq (1), slide 26,

$$\sum_{j=1}^n \frac{\partial \log f(Y_j | X_j; \beta)}{\partial \beta} = \sum_{i=1}^n \begin{bmatrix} 1 \\ X_j \end{bmatrix} \left\{ Y_j - \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)} \right\} = \begin{bmatrix} \sum_{j=1}^n \left\{ Y_j - \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)} \right\} \\ \sum_{j=1}^n X_j \left\{ Y_j - \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)} \right\} \end{bmatrix}$$

. Then,

$$\frac{\partial^2 \sum_{i=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T} = - \sum_{j=1}^n \begin{bmatrix} 1 \\ X_j \end{bmatrix} p_j (1 - p_j) \begin{bmatrix} 1 & X_j \end{bmatrix} = - \sum_{i=1}^n \begin{bmatrix} p_j (1 - p_j) & X_j p_j (1 - p_j) \\ X_j^2 p_j (1 - p_j) \end{bmatrix}$$

where, as previously defined, $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$. Since $\frac{\partial^2 \sum_{i=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T}$ is not a function of Y_i ,

$$-E(\mathbf{Y} | \mathbf{X}) \left\{ \frac{\partial^2 \sum_{j=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T} \right\} = - \frac{\partial^2 \sum_{j=1}^n \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T}.$$

1. Wald test

- After simplification and taking the inverse of the 2X2 expected information matrix, we have

$$\text{Var}(\hat{\beta}^*) = \frac{\sum_{j=1}^n p_j(1 - p_j)}{\left\{ \sum_{i=1}^n p_i(1 - p_i) \right\} \left\{ \sum_{j=1}^n X_j^2 p_j(1 - p_j) \right\} - \left\{ \sum_{i=1}^n X_i p_i(1 - p_i) \right\}^2}$$

- And

$$\widehat{\text{Var}}(\hat{\beta}^*) = \frac{\sum_{j=1}^n \hat{p}_j(1 - \hat{p}_j)}{\left\{ \sum_{j=1}^n \hat{p}_j(1 - \hat{p}_j) \right\} \left\{ \sum_{j=1}^n X_j^2 \hat{p}_j(1 - \hat{p}_j) \right\} - \left\{ \sum_{j=1}^n X_j \hat{p}_j(1 - \hat{p}_j) \right\}^2} \quad (2)$$

$$\text{where } \hat{p}_j = \frac{\exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}.$$

Example: HPTN 037

- The HIV Prevention Trials Network (HPTN) 037 Study evaluated a peer-driven intervention to promote HIV risk reduction among people who inject drugs (PWID) and their injection drug and sexual network members in Philadelphia, US.
- PWID volunteered as indexes, served as the initial point of contact into their HIV risk network, and recruited other immediate HIV risk network members, which formed non-overlapping egocentric networks around each index member.
- Eligible indexes and their risk networks were randomized to the experimental intervention or control. Follow-up visits occurred at six-month intervals following randomization for up to 30 months.
- 232 indexes and 464 network members were examined to assess changes in injection risk behaviors.

Example: HPTN 037

- In this example, the primary outcome is a cumulative 30-months risk injection behavior, “**shared cotton**”, which is an indicator variable for sharing needle/syringe “works”.
- $Y_j = 1$ for the participants having the risk injection behavior, $Y_j = 0$ for otherwise.
- $X_j = 1$ for the participant in the intervention group, $X_j = 0$ for the participant is in control group.
- total $n = 696$ participants
- Logistic regression is used to estimate the overall effect of the intervention:

$$f(Y_j|X_j) = \frac{\exp(\beta_0 + \beta^* X_j)^{Y_j}}{1 + \exp(\beta_0 + \beta^* X_j)}$$

- We are interested in testing $H_0 : \beta^* = 0$ vs. $H_0 : \beta^* \neq 0$.

Example: HPTN 037

See "Three Hypothesis Tests of HPTN.R" in Files/Lecture Notes/Week 1/R Code for the hard coding of the three test statistics derived from first principles below

- We first estimate the parameters using GEE:

```
fit1 <- geeglm( cotton ~ Treat, data = base_hptn_30 ,  
              id=uid , family = binomial(link="logit") )
```

and we obtain that $\hat{\beta} = (-0.022, -0.314)$,

- Then $\widehat{Var}(\hat{\beta}^*)$ can be calculated by (2), which is 0.0234 (SE = 0.153);
- $Z^2 = \frac{(\hat{\beta}^*)^2}{\widehat{Var}(\hat{\beta}^*)} = 4.228$;
- Given $\alpha = 0.05$, p-value = $P(Z^2 > 4.228) = 0.0397 < 0.05$
- Thus, we reject the H_0 given 5% significance level.
- The result can be checked by using

```
> summary(fit1)  
  
Call:  
geeglm(formula = cotton ~ Treat, family = binomial(link = "logit"),  
       data = base_hptn_30, id = uid)  
  
Coefficients:  
              Estimate Std.err Wald Pr(>|W|)  
(Intercept)  -0.0222   0.1054  0.04    0.83  
Treat         -0.3142   0.1528  4.23    0.04 *  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2. Score test

- The first derivative of the log likelihood is often referred to as the score function or score statistic, and denoted U_j .

$$U_j = U_j(Y_j, X_j; \beta) = \frac{\partial \log f(Y_j | X_j; \beta)}{\partial \beta} \text{ and } \mathbf{U} = \sum_{j=1}^n U_j(Y_j, X_j; \beta).$$

- Note that by definition of the MLE, $\sum_{j=1}^n U_j(\hat{\beta}) = \mathbf{0}$ since the MLE, $\hat{\beta}$, is the solution to this equation, where, here, $\dim(U_j) = 2 \times 1$ corresponding to $(\beta_0, \beta^*)^T$.

- In addition, for GLMs such as logistic and linear regression, U_j has the form $w_j(X_j; \beta)[Y_j - E(Y_j | X_j; \beta)]$, guaranteeing that

$$\begin{aligned} E_{Y|X}[w_j(X_j; \beta)(Y_j - E(Y_j | X_j; \beta))] &= \\ w_j(X_j; \beta)[E_{Y|X}(Y_j | X_j; \beta) - E(Y_j | X_j; \beta)] &= 0 \end{aligned}$$

- In general, if $\hat{\beta}$ solves unbiased estimating equations such as the above, $\hat{\beta}$ is consistent for β (Rao 1965*, Huber 1967†, Liang and Zeger 1986‡)

* Rao, Calyampudi Radhakrishna, et al. Linear statistical inference and its applications. Vol. 2. New York: Wiley, 1973.

† Huber, Peter J. "UNDER NONSTANDARD CONDITIONS." Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability 1 (1967): 21-37.

‡ Liang, Kung-Yee, and Scott L. Zeger. "Longitudinal data analysis using generalized linear models." Biometrika 73.1 (1986): 13-22.

2. Score test

- It has been shown that

$$\text{Var}(U_j) = -E_{Y_j|X_j} \left\{ \frac{\partial^2 f_j(Y_j, X_j; \beta)}{\partial \beta \partial \beta^T} \right\} = E_{Y_j|X_j}(U_j U_j^T),$$

and we denoted $\text{Var}(\mathbf{U}) = \sum_{j=1}^n \text{Var}(U_j)$ as $\mathbf{I}$. Note that either expression can be used to estimate the variance.

- Thus, the score test statistic for $H_0 : \beta^* = 0$ vs. $H_A : \beta^* \neq 0$ takes the value of $\mathbf{U}^T \mathbf{I}^{-1} \mathbf{U}$ as $Z^2 \xrightarrow{d} \chi_1^2$ under H_0 , where

$$\mathbf{U}(\beta_{H_0}) = \sum_{j=1}^n U_j(X_j; \beta^* = 0, \beta_{-r}),$$

β_{-r} is the $(r-1) \times 1$ vector of β where β^* , the r th element, is deleted. And

$$\mathbf{I}(\beta_{H_0}) = - \sum_{j=1}^n E_{Y_j|X_j} \left[\frac{\partial^2 \log f(Y_j|X_j; \beta^* = 0, \beta_{-r})}{\partial \beta \partial \beta^T} \right]$$

Example: Logistic regression

- Recall that $f(Y_j|X_j) = \frac{\exp(\beta_0 + \beta^* X_j)^{Y_j}}{1 + \exp(\beta_0 + \beta^* X_j)}$ and $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$
- The log-likelihood for the study with data $\{\mathbf{Y}, \mathbf{X}\}$ of sample size n ,

$$L(\beta) = \sum_{j=1}^n Y_j(\beta_0 + \beta^* X_j) - \log\{1 + \exp(\beta_0 + \beta^* X_j)\}$$

Noting that in two parameter GLMs as here, $\hat{\beta}_{H_0} = \hat{\beta}_{0|H_0}$

$$\begin{aligned} U(\hat{\beta}_{H_0}) &= \sum_{j=1}^n \frac{\partial \log f(Y_j|X_j; \beta^* = 0, \hat{\beta}_{H_0})}{\partial \beta} = \sum_{i=1}^n \begin{bmatrix} 1 \\ X_j \end{bmatrix} \left\{ Y_j - \frac{\exp(\hat{\beta}_{H_0})}{1 + \exp(\hat{\beta}_{H_0})} \right\} \\ &= \begin{bmatrix} \sum_{j=1}^n \left\{ Y_j - \frac{\exp(\hat{\beta}_{H_0})}{1 + \exp(\hat{\beta}_{H_0})} \right\} \\ \sum_{j=1}^n X_j \left\{ Y_j - \frac{\exp(\hat{\beta}_{H_0})}{1 + \exp(\hat{\beta}_{H_0})} \right\} \end{bmatrix} \end{aligned} \quad (3)$$

Example: Logistic regression

$$I(\hat{\beta}_{H_0}) = \sum_{j=1}^n \begin{bmatrix} \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\} & X_j \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\} \\ X_j \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\} & X_j^2 \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\} \end{bmatrix}. \quad (4)$$

Then, the score test statistic, Z^2 , is

$$Z^2 = \frac{1}{Z_3} \left[\left(\sum_{j=1}^n Z_{1j} \right) \left(\sum_{j=1}^n Z_{2j} \right)^2 - 2 \left(\sum_{j=1}^n Z_{2j} \right) \left(\sum_{j=1}^n X_j Z_{1j} \right) \left(\sum_{j=1}^n X_j Z_{2j} \right) + \left(\sum_{j=1}^n X_j^2 Z_{1j} \right) \left(\sum_{j=1}^n X_j Z_{2j} \right)^2 \right] \quad (5)$$

where $Z_{1j} = \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \left\{ 1 - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})} \right\}$, $Z_{2j} = Y_j - \frac{\exp(\hat{\beta}_{H_0})}{1+\exp(\hat{\beta}_{H_0})}$, and

$$Z_3 = \left(\sum_{j=1}^n Z_{1j} \right) \left(\sum_{j=1}^n X_j^2 Z_{1j} \right) - \left(\sum_{j=1}^n X_j Z_{1j} \right)^2.$$

Example: HPTN 037

- We first estimate the parameter using GLM based on the model under H_0 : $f(Y_j|X_j, H_0) = \frac{\exp(\beta_0)Y_j}{1+\exp(\beta_0)}$, we found that $\hat{\beta}_{H_0} = \hat{\beta}_{0|H_0} = -0.173$.
- We obtained $\hat{\beta}_{0|H_0}$ by fitting the null model to the data:

```
> glm.fit0 <- glm(cotton~1, data = base_hptn_30, family = binomial(link="logit"))  
> fit0=glm.fit0  
> summary(fit0)
```

Call:

```
glm(formula = cotton ~ 1, family = binomial(link = "logit"),  
    data = base_hptn_30)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.105	-1.105	-1.105	1.252	1.252

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.17284	0.07609	-2.271	0.0231 *

Example: HPTN 037

- Then $\mathbf{U}(\hat{\beta}_{H_0})$ and $\mathbf{I}(\hat{\beta}_{H_0})$ can be calculated using (3) and (4)
From "Three Hypothesis Tests of HPTN.R" in Files/Lecture Notes/Week 1/R Code we obtain
- $Z^2 = \mathbf{U}(\hat{\beta}_{H_0})^T \mathbf{I}(\hat{\beta}_{H_0})^{-1} \mathbf{U}(\hat{\beta}_{H_0}) = 4.237$
- Given $\alpha = 0.05$, $p\text{-value} = P(Z^2 > 4.237) = 0.040 < 0.05$
- Thus, we reject the H_0 given 5% significance level.
- The result can be checked by using

```
> fit2 <- glm(cotton~ Treat, data = base_hptn_30, family = binomial(link="logit"))
> anova(fit2, test ="Rao")
Analysis of Deviance Table

Model: binomial, link: logit
Response: cotton
Terms added sequentially (first to last)

      Df Deviance Resid. Df Resid. Dev  Rao Pr(>Chi)
NULL                                695      960
Treat  1      4.24      694      955  4.24    0.04 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3. The Likelihood ratio test (LRT)

- The test statistic Z^2 is

$$\begin{aligned} Z^2 = & -2\{\log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \beta_{-r} = \hat{\beta}_{-r}) \\ & - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = \hat{\beta}^*, \beta_{-r} = \hat{\beta}_{-r})\} \\ & \xrightarrow{d} \chi_1^2 \end{aligned}$$

Example: HPTN 037

- As with the score test, we estimate the parameter based on the model under H_0 : $f(Y_j|X_j; \beta_{H_0}) = \frac{\exp(\beta_0) Y_j}{1 + \exp(\beta_0)}$, we obtain $\hat{\beta}_0|_{H_0} = \hat{\beta}_0|_{H_0} = -0.173$,

Example: HPTN 037

- Then we estimate the parameter based on the model under H_A ,
 $f(Y_j|X_j, H_A) = \frac{\exp(\beta_0 + \beta^* X_j)^{Y_j}}{1 + \exp(\beta_0 + \beta^* X_j)}$, we obtain that $\hat{\beta} = (-0.022, -0.314)$.
From "Three Hypothesis Tests of HPTN.R" in Files/Lecture Notes/Week 1/R Code we obtain
- $Z^2 = -2\{\log f(\mathbf{Y}|\mathbf{X}, \hat{\beta}_0 = -0.173, \beta^* = 0) - \log f(\mathbf{Y}|\mathbf{X}, \hat{\beta}_0 = -0.022, \hat{\beta}^* = -0.314)\} = 4.242$
- Given $\alpha = 0.05$, p-value $= P(Z^2 > 4.242) = 0.0394 < 0.05$
- Thus, we reject the H_0 .
- The result can be checked by using

```
> anova(fit2, test = "LRT")
Analysis of Deviance Table

Model: binomial, link: logit

Response: cotton

Terms added sequentially (first to last)
```

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL			695	960	
Treat	1	4.24	694	955	0.039 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Comparison of the three tests

Table 1: Hypothesis Tests Results of HPTN 037 data

Tests	Test Statistics (Z^2)	P-value
Wald Test	4.228	0.0397
Score Test	4.237	0.0400
LRT	4.242	0.0394

- It has been found that the LRT has better finite sample properties than the score test, which has better finite sample properties than the Wald test. (Vaeth *)
- For power calculations, the expected value of the score test under the null and alternative is often difficult to calculate, and the Wald test is most commonly used, then the LRT. For semi-parametric inference, as in GEEs, the LRT is not available

* Vaeth M. On the use of Wald's test in exponential families. International Statistical Review 1985; 53:199-214

- Now that we have test statistics, we can consider power calculations.
- The power of the study is the probability that the null hypothesis will be rejected when it is false.
- The power of the study is calculated for a pre-specified value of β^* , the intervention effect, also known as the parameter of interest.
- The problem is set up so that the test statistic rejects the null $1 - \alpha/2$ (for two-sided test) when the null hypothesis is true
- $\text{Power}(\pi)$: $\pi = \Pr(Z^2 > c | H_A) = 1 - \Pr(Z^2 \leq c | H_A) = 1 - \Pr(Z_\lambda^2 \leq 3.84)$, where $c = 3.84$ is the critical value of a central χ_1^2 for $\alpha = 5\%$, λ is the non-centrality parameter, a function of the alternative value of the effect estimate, $\beta_{H_A}^*$, of n , and other aspects of the design that determine $\text{Var}_{H_A}(\hat{\beta}_{H_A}^*)$, when the Wald test is being used, and otherwise a more complex function of n and other design features when the score test or LRT is being used for power calculations. Z_λ^2 is a random variable following the non-central $\chi_1^2(\lambda)$ distribution with 1 degree of freedom and non-centrality parameter λ .

Sample size and Minimum detectable effect (MDE)

- **Sample size**, n , solves $\pi = 1 - Pr(Z_{\lambda(n)}^2 \leq 3.84)$, where $\lambda(n)$ is a function of n through the variance of the estimator of the intervention effect (Wald test), and otherwise a more complex function of n (score test, LRT) and π is pre-specified power, usually 80%.
- **Minimum detectable effect size (MDE)** Here, power and sample size are fixed.
 - This is often the case in grant proposals because the budget is limited.
 - Hence, with a given budget/sample size and the usual 80% power constraint, we want to know what our minimum detectable effect size is, that is,
 - find β^* that solves $\pi = 1 - Pr(Z_{\lambda(\beta^*)}^2 \leq 3.84)$, where $\lambda(\beta^*)$, the non-centrality parameter, is a function of β^* both in its numerator and in the denominator, at least for categorical/binomial/Bernoulli data.

Using the normal distribution instead of χ^2 for study design

It should be noted that instead of using the central and non-central χ^2 distributions for power, sample size and MDE calculations for study design, the normal distribution can be used.

Here, $\pi = Pr(|Z| > c_\phi | H_A) = Pr(Z > c_\phi | H_A) + Pr(-Z < -c_\phi | H_A)$, where c_ϕ is the 97.5th percentile of the standard normal (0,1) distribution, 1.96.

This approach can also be used for power, sample size and MDE calculations.

Calculating the non-centrality parameter λ

- Given required power π , significance level α , and degrees of freedom of the χ^2 distribution, we can obtain the $\lambda(\pi)$ by solving

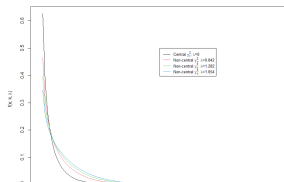
$$\pi = 1 - \int_0^{3.84} f(x; 1, \lambda(\pi)) dx, \quad (6)$$

for $\lambda(\pi)$, where $f(x; 1, \lambda)$ is the probability density function of the noncentral χ^2 distribution with DF=1 and

- non-centrality parameter λ .
For example, given $\pi = 80\%$, $\alpha = 5\%$ and DF=1, we have $\lambda(\pi) = 7.85$, when $\pi = 90\%$, $\lambda(\pi) = 10.5$

```
fx=function(x,lambda){  
  dchisq(x, 2, ncp = lambda, log = FALSE)  
}  
  
power.eq1 <-function(lambda){  
  # 0.8 = 1-int^3.84_0 chi^2(1,lambda) => 0 = int^3.84_0 chi^2(1,lambda)-0.2  
  power = integrate(fx, 0, 3.84, lambda=lambda)$value-0.2  
}  
find.para<-multiroot(power.eq1, start =10, positive = TRUE)
```

Probability density function of χ^2_1 with non centrality parameter λ



Three types of power analysis

- To find the power, π , for a study with sample size n assuming $\beta = \beta^*$, we calculate $\pi = 1 - \int_0^{3.84} f(x; 1, \lambda(n, \beta^*)) dx$. Here $\lambda(n, \beta^*)$ is a function of n and β^* and is the parameter indexing a non-central χ^2 distribution with 1 degree of freedom.
- For sample size and MDE calculations, we solve for the design parameter that yields a non-centrality parameter indexing the non-central χ^2 for which one minus its cumulative distribution function to 3.84 has probability π .
 - When the required power is $\pi = 80\%$ at 5% significance level, we obtain n using $\lambda(\pi = 80\%) = 7.85$; when the required power is $\pi = 90\%$ at 5% significance level, we obtain n using $\lambda(\pi = 90\%) = 10.5$;
 - When the required power is $\pi = 80\%$ at 5% significance level, we obtain MDE (β^*) using $\lambda(\pi = 80\%) = 7.85$; When required power $\pi = 90\%$ at 5% significant level, we obtain MDE (β^*) using $\lambda(\pi = 80\%) = 10.5$.

Power Analysis based upon the Wald test

- Wald Test:

- the non-centrality parameter $\lambda(\pi) = n \times \frac{(\beta^*)^2}{I_j^{-1}(\hat{\beta}^*; \beta_{0,H_A}, \beta_{H_A}^*)}$ with $I_j^{-1}(\hat{\beta}^*; \beta_{0,H_A}, \beta_{H_A}^*)$ is the $r - r$ th element the covariance matrix $Var(\hat{\beta})$ of $\hat{\beta}$ based on the expected information matrix for a single observation $I_j(\hat{\beta}; \beta_{0,H_A}, \beta_{H_A}^*)$, where $I_j(\hat{\beta})$ is evaluated at $\beta = \beta_{H_A} = (\beta_{0,H_A}, \beta_{H_A}^*)$.
- Sample size n can be obtained by solving $n = \lambda(\pi) \times \frac{I_j^{-1}(\hat{\beta}^*; \beta_{0,H_A}, \beta_{H_A}^*)}{(\beta^*)^2}$
- MDE of β^* can be obtained by solving $\lambda(\pi) = n \times \frac{(\beta^*)^2}{I_j^{-1}(\hat{\beta}^*; \beta_{0,H_A}, \beta_{H_A}^*)}$ given n , π , α and β_0 , for β^* .

Power Analysis based upon the Wald test

- In the logistic regression example:

- $X \sim \text{Bern}(p_x)$ and $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$

-

$$I_j(\hat{\beta}) = E_X \left[E_{Y|X} \left\{ - \frac{\partial^2 \log f(Y_j | X_j; \beta)}{\partial \beta \partial \beta^T} \bigg|_{\beta = (\beta_0, H_A, \beta_{H_A}^*)} \right\} \right]$$
$$= \begin{bmatrix} I_{j, \beta_0} & I_{j, \beta_0 \beta^*} \\ I_{j, \beta_0 \beta^*} & I_{j, \beta^*} \end{bmatrix}$$

where

$$I_{j, \beta_0} = E_X \left[\frac{\exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^* X_j)\}^2} \right] =$$

$$p_x \frac{\exp(\hat{\beta}_0 + \hat{\beta}^*)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^*)\}^2} + (1 - p_x) \frac{\exp(\hat{\beta}_0)}{\{1 + \exp(\hat{\beta}_0)\}^2},$$

$$I_{j, \beta^*} = E_X \left[\frac{\exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^* X_j)\}^2} X_j^2 \right] = p_x \frac{\exp(\hat{\beta}_0 + \hat{\beta}^*)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^*)\}^2}$$

$$I_{j, \beta_0 \beta^*} = E_X \left[\frac{\exp(\hat{\beta}_0 + \hat{\beta}^* X_j)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^* X_j)\}^2} X_j \right] = p_x \frac{\exp(\hat{\beta}_0 + \hat{\beta}^*)}{\{1 + \exp(\hat{\beta}_0 + \hat{\beta}^*)\}^2}.$$

- If pilot data are available, $I_{j, \beta_0 \beta^*}$ can be evaluated at $\beta = (\beta_0, H_A, \beta_{H_A}^*)$

Power Analysis based upon the Wald test

- $Var(\hat{\beta}) = I_j^{-1}(\hat{\beta})$, which is a 2 by 2 matrix inversion

- The 2-2th element of $I_j^{-1}(\hat{\beta})$ is

$$I_j^{-1}(\hat{\beta}^*) = \frac{1}{1-p_x} \frac{\{1+\exp(\hat{\beta}_0)\}^2}{\exp(\hat{\beta}_0)} + \frac{1}{p_x} \frac{\{1+\exp(\hat{\beta}_0+\hat{\beta}^*)\}^2}{\exp(\hat{\beta}_0+\hat{\beta}^*)}$$

Power Analysis based upon the Score test

- β_{H_0} is the β_0 and β^* under H_0 , where often $\beta^* = 0$, and $\beta_{H_A} = (\beta_{0|H_A}, \beta^*)^T$ under H_A . Often, $\beta_{0|H_0} = \beta_{0|H_A}$. This is true, in general, in trials of new interventions compared to standard of care, where the outcome rates are those to be expected in the absence of the intervention, and to change as a result of the intervention if it is effective.
- We denote the r th element in the score function $\mathbf{U}$ as U_{β^*} for β^* , and the score function for β_0 as U_{β_0} .
- The score test:
 - The test statistic is computed as the quadratic form $\mathbf{U}^T \mathbf{I}^{-1} \mathbf{U}$ under H_0

Power Analysis based upon the score test

- To conduct power analysis based upon the score test, the expected value of the non-centrality parameter, λ must be derived as a function of n , β^* and other design parameters, such as $E(Y)$ and $Pr(X)$
- It has been shown that the non-centrality parameter, $\lambda = \xi^2 / I^*(\hat{\beta}^*)$, where ξ is the expectation of

$$U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) - I_{\beta_0 \beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) I_{\beta_0 \beta_0}^{-1}(\tilde{\beta}_0, \beta_{H_0}^*) U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*) \quad (7)$$

and $I^*(\hat{\beta}^*)$ is the variance of (7) with respect to $(\mathbf{Y}, \mathbf{X})$ evaluated at $\tilde{\beta}_0$ and $\beta_{H_0}^*$.

- What is $\tilde{\beta}_0$? It is the expected value of β_0 with respect to $(\mathbf{Y}, \mathbf{X})$ under H_A from the fit of the mis-specified null model, where $\beta^* = 0$ (if that is the null)
- The general formula for λ for generalized linear models (GLMs) is given by Self (1988)[§]
- As always, sample size n can be obtained using the non-centrality parameter, here $\lambda(\pi) = \xi^2 / I^*(\beta^*)$ given $\lambda(\pi)$, $\tilde{\beta}_0$, and $\beta_{H_0}^*$.
- MDE β^* can be solved using $\lambda(\pi) = \xi^2 / I^*(\hat{\beta}^*)$ given n , $\lambda(\pi)$ and β_0 at H_0 , β_0 at H_A .

[§] Self, Steven G., and Robert H. Mauritsen. "Power/sample size calculations for generalized linear models." *Biometrics* (1988): 121-138.

The score test: Logistic Regression

Now, we need to find $\xi^2 = E_{Y,X}[U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1} U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)]$ and $I^*(\hat{\beta}^*) = \text{Var}\{U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1} U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)\}$ with respect to (Y, X) evaluated at $\tilde{\beta}_0$ and $\beta_{H_0}^* = 0$.

- First, we find $\tilde{\beta}_0$, then $U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*)$, then $I_{\beta_0\beta^*}$ and then $I_{\beta_0\beta_0}$. We put them all together to obtain λ .

$\tilde{\beta}_0$, is the solution of $\mathbf{0} = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbf{U}_{\beta_0}(\beta_{0|H_A}, \beta_{H_0}^*)$, and its estimator is $\hat{\tilde{\beta}}_0$.

- With $X \sim \text{Bern}(p_x)$, and $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$, based on (1)
- $Pr(Y = 1|X = 0) = p_0 = \frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})}$ and
 $Pr(Y = 1|X = 1) = p_1 = \frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}$
- $p^* = (1 - p_x)p_0 + p_x p_1$ and $\tilde{\beta}_0$ is *logit*(p^*) (Liu and Liang, 1997*).

[†] Liu, Guanghan, and Kung-Yee Liang. "Sample size calculations for studies with correlated observations." *Biometrics* (1997): 937-947.

The score test: Logistic Regression

- Next we find $U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*)$ and $U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)$

$$U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) = \sum_{j=1}^n X_i \left\{ Y_i - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\} = \sum_{j=1}^n X_i (Y_i - p^*)$$

and

$$U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*) = \sum_{j=1}^n \left\{ Y_i - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\} = \sum_{j=1}^n (Y_i - p^*)$$

The score test: Logistic Regression

- Next we find $I_{\beta_0\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*)$, $I_{\beta_0\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)$:

$$\begin{aligned} I_{\beta_0\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) &= E \left\{ \frac{\partial^2 p_i}{\partial \beta^* \beta_0} \right\} \bigg|_{\beta_{H_0}^*, \tilde{\beta}_0} = E_X[E_{Y|X} \{X_i p_j(1 - p_j)\}] \big|_{\beta_{H_0}^*, \tilde{\beta}_0} \\ &= p_x \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \left\{ 1 - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\} \\ &= p_x p^*(1 - p^*) \end{aligned} \quad (8)$$

and

$$\begin{aligned} I_{\beta_0\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*) &= E \left\{ \frac{\partial^2 p_i}{\partial \beta_0^2} \right\} \bigg|_{\beta_{H_0}^*, \tilde{\beta}_0} = E_X[E_{Y|X} \{p_j(1 - p_j)\}] \big|_{\beta_{H_0}^*, \tilde{\beta}_0} \\ &= \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \left\{ 1 - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\} \\ &= p^*(1 - p^*) \end{aligned} \quad (9)$$

The score test: Logistic Regression

- Then, the numerator of the non-centrality parameter,

$$\begin{aligned}\xi &= E_{Y,X}[U_{\beta^*}(\tilde{\beta}_0, \beta_{H_0}^*) - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1} U_{\beta_0}(\tilde{\beta}_0, \beta_{H_0}^*)] \\&= E_{X,Y} \sum_{j=1}^n (Y_i - p^*)(X_i - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1}) \\&= n E_X[E_{Y|X}\{(Y_i - p^*)(X_i - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1})\}] \\&= n E_X[(p_i - p^*)(X_i - I_{\beta_0\beta^*} I_{\beta_0\beta_0}^{-1})] \\&= n [p_x\{(p_1 - p^*)(1 - p_x)\} - (1 - p_x)\{(p_0 - p^*)p_x\}] \\&= n * p_x(1 - p_x)(p_1 - p_0)\end{aligned}$$

where $p^* = (1 - p_x)p_0 + p_x p_1$, $I_{\beta_0\beta_0}$ is given by (8), and $I_{\beta_0\beta^*}$ is given by (9).

The score test: Logistic Regression

- The denominator of the non-centrality parameter,

$$\begin{aligned} I_{\beta^*}^* &= nE_X\{p_i(1-p_i)(X_i - I_{\beta_0\beta^*}I_{\beta_0\beta_0}^{-1})^2\} \\ &= n[p_x\{p_1(1-p_1)(1-p_x)^2\} + p_0(1-p_0)(1-p_x)(-p_x)^2] \\ &= np_x(1-p_x)\{(1-p_x)p_1(1-p_1) + p_xp_0(1-p_0)\} \end{aligned}$$

- As a result, we have

$$\begin{aligned} \lambda(\pi) &= \xi^2 / I^*(\beta^*) = \frac{\{n \times p_x(1-p_x)(p_1-p_0)\}^2}{np_x(1-p_x)\{(1-p_x)p_1(1-p_1) + p_xp_0(1-p_0)\}} \\ &= n \frac{(p_1-p_0)^2 p_x(1-p_x)}{(1-p_x)p_1(1-p_1) + p_xp_0(1-p_0)} \end{aligned} \quad (10)$$

- For sample size n , we have $n = \frac{\lambda(\pi) \times (1-p_x)p_1(1-p_1) + p_xp_0(1-p_0)}{(p_1-p_0)^2 p_x(1-p_x)}$ with $\lambda(80\%) = 7.85$ and $\lambda(90\%) = 10.5$ at 5% significance level.
- For MDE, we solve (10) given p_x , $\beta_{0|H_A}$, $\lambda(\pi)$ and n .

Power Analysis based upon the LRT

- Similar to the score test, we need to approximate the distribution of the test statistic

$$Z^2 = 2\{\log f(\mathbf{Y}|\mathbf{X}; \hat{\beta}_{H_A}) - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \hat{\beta}_{-r|H_0})\}$$

under H_A (Self, 1992[¶]).

- Self suggests a decomposition of Z^2 into three terms:

$$\begin{aligned} Z^2 = & 2\{\log f(\mathbf{Y}|\mathbf{X}; \hat{\beta}_{H_A}) - \log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})\} \\ & - 2\{\log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \hat{\beta}_{-r|H_0}) - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \tilde{\beta}_{-r})\} \\ & + 2\{\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})\} - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \tilde{\beta}_{-r})\} \end{aligned} \quad (11)$$

where $\tilde{\beta}_{-r}$ can be obtained as described in page 52 for the score test, and its estimator is $\hat{\tilde{\beta}}_{-r}$.

- When $\beta = (\beta_0, \beta^*)$, $\tilde{\beta}_{-r} = \tilde{\beta}_0$ and $\hat{\beta}_{0|H_0}$ is the estimator of $\tilde{\beta}_0$.
- Self shows that an approximate expected value for the first term in (11) is the number of parameters in the model. In the case of $\beta = (\beta_0, \beta^*)$, it takes value of 2.



Self, Steven G., Robert H. Mauritsen, and Jill Ohara. "Power calculations for likelihood ratio tests in generalized linear models." *Biometrics* (1992): 31-39.

Power Analysis based upon the LRT

- An approximation of the expected value of the second term in (11) equals to a **trace** of the matrix M , where

$$M = \left[E_{\mathbf{Y}\mathbf{X}} \left\{ \frac{\partial^2 \log f(\mathbf{Y}|\mathbf{X}; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}_{-r} \partial \boldsymbol{\beta}_{-r}^T} \middle|_{(\tilde{\boldsymbol{\beta}}_{-r}, \boldsymbol{\beta}_{H_0}^*)} \right\} \right]^{-1} E_{\mathbf{Y}\mathbf{X}} \left[\frac{\partial \log f(\mathbf{Y}|\mathbf{X}; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}_{-r}} \middle|_{(\tilde{\boldsymbol{\beta}}_{-r}, \boldsymbol{\beta}_{H_0}^*)} \right]^{\otimes 2},$$

where for a vector $\mathbf{a}$, $\mathbf{a}^{\otimes 2} = \mathbf{a}\mathbf{a}^T$.

- The first term of M is the expectation of the second derivative of log-likelihood function evaluated at $(\tilde{\boldsymbol{\beta}}_{-r}, \boldsymbol{\beta}_{H_0}^*)$.
- The second term of M is the variance of the first derivative of log-likelihood function evaluated at $(\tilde{\boldsymbol{\beta}}_{-r}, \boldsymbol{\beta}_{H_0}^*)$.
- Under the setting with $\boldsymbol{\beta} = (\boldsymbol{\beta}_0, \boldsymbol{\beta}^*)^T$, M is a scalar and $\text{trace}(M) = M$.

Power Analysis based upon the LRT

- The expected value of the third term in (11), denoted by Δ , can be calculated exactly:

$$\Delta = E_{\mathbf{Y}|\mathbf{X}}[2\{\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})\} - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \tilde{\beta}_{-r})\}]$$

- The expected value of a noncentral χ^2 random variable with 1 degree of freedom and noncentrality parameter λ is $1 + \lambda$.
- Equating this to the approximation derived from (11), we can obtain

$$\lambda(\pi) = \Delta + 1 - \text{trace}(M)$$

- Under the setting with $\beta = (\beta_0, \beta^*)^T$,

$$\lambda(\pi) = \Delta + 1 - M,$$

which we can use to calculate power or solve for n or MDE.

LRT: Logistic Regression Example

- $X \sim \text{Bern}(p_x)$ and $p_j = \frac{\exp(\beta_0 + \beta^* X_j)}{1 + \exp(\beta_0 + \beta^* X_j)}$
- $Pr(Y = 1|X = 0) = p_0 = \frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})}$ and
 $Pr(Y = 1|X = 1) = p_1 = \frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}$
- $p^* = (1 - p_x)p_0 + p_x p_1$ and $\tilde{\beta}_0$ is *logit*(p^*)
- To find λ , we need to calculate Δ and $\text{trace}(M) = M$.

1. We first calculate

$\Delta = E_{\mathbf{Y}|\mathbf{X}}[2\{\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})\} - \log f(\mathbf{Y}|\mathbf{X}; \beta^* = 0, \tilde{\beta}_0)]$ by noting that

$$\begin{aligned} E_{\mathbf{Y}|\mathbf{X}}[\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})] &= E_{\mathbf{X}}[E_{\mathbf{Y}|\mathbf{X}}[\log f(\mathbf{Y}|\mathbf{X}; \beta_{H_A})]] \\ &= n E_{\mathbf{X}} \left[\frac{\exp(\beta_{0|H_A} + \beta^* X_j)}{1 + \exp(\beta_{0|H_A} + \beta^* X_j)} (\beta_{0|H_A} + \beta^* X_j) - \log\{1 + \exp(\beta_{0|H_A} + \beta^* X_j)\} \right] \\ &= n p_x \left[\frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)} (\beta_{0|H_A} + \beta^*) - \log\{1 + \exp(\beta_{0|H_A} + \beta^*)\} \right] \\ &\quad + n(1 - p_x) \left[\frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})} (\beta_{0|H_A}) - \log\{1 + \exp(\beta_{0|H_A})\} \right] \end{aligned}$$

LRT: Logistic Regression Example

$$\begin{aligned} E_{\mathbf{Y}|\mathbf{X}}[\log f(\mathbf{Y}|\mathbf{X}; \tilde{\beta}_0, \beta^* = 0)] &= E_{\mathbf{X}}[E_{\mathbf{Y}|\mathbf{X}}[\log f(\mathbf{Y}|\mathbf{X}; \tilde{\beta}_0, \beta^* = 0)]] \\ &= nE_{\mathbf{X}}\left[\frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}(\tilde{\beta}_0) - \log\{1 + \exp(\tilde{\beta}_0)\}\right] \\ &= np_x\left[\frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}(\tilde{\beta}_0) - \log\{1 + \exp(\tilde{\beta}_0)\}\right] \\ &\quad + n(1 - p_x)\left[\frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}(\tilde{\beta}_0) - \log\{1 + \exp(\tilde{\beta}_0)\}\right] \end{aligned}$$

Then we have

$$\begin{aligned} \Delta &= 2n\left[p_x\left\{\frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}(\beta_{0|H_A} + \beta^*) - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}\tilde{\beta}_0\right.\right. \\ &\quad \left.- \log\left(\frac{1 + \exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\tilde{\beta}_0)}\right)\right\} + (1 - p_x)\left\{\frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})}\beta_{0|H_A}\right. \\ &\quad \left.- \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}\tilde{\beta}_0 - \log\left(\frac{1 + \exp(\beta_{0|H_A})}{1 + \exp(\tilde{\beta}_0)}\right)\right\}\right] \end{aligned}$$

LRT: Logistic Regression Example

Δ can be further simplified to

$$\begin{aligned}\Delta &= 2n \left[p_x \left\{ p_1(\beta_{0|H_A} + \beta^* - \tilde{\beta}_0) - \log \left(\frac{1 + \exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right. \\ &\quad \left. + (1 - p_x) \left\{ p_0(\beta_{0|H_A} - \tilde{\beta}_0) - \log \left(\frac{1 + \exp(\beta_{0|H_A})}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right] \\ &= n\Delta^*,\end{aligned}\tag{12}$$

where

$$\begin{aligned}\Delta^* &= 2 \left[p_x \left\{ p_1(\beta_{0|H_A} + \beta^* - \tilde{\beta}_0) - \log \left(\frac{1 + \exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right. \\ &\quad \left. + (1 - p_x) \left\{ p_0(\beta_{0|H_A} - \tilde{\beta}_0) - \log \left(\frac{1 + \exp(\beta_{0|H_A})}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right]\end{aligned}$$

LRT: Logistic Regression Example

2. Next, we can obtain

$$M = E_{\mathbf{Y}\mathbf{X}} \left[- \frac{\partial^2 \log f(\mathbf{Y}|\mathbf{X};\beta)}{\partial \beta_0^2} \Big|_{(\tilde{\beta}_0, \beta_{H_0}^*)} \right]^{-1} E_{\mathbf{Y}\mathbf{X}} \left[\left\{ \frac{\partial \log f(\mathbf{Y}|\mathbf{X};\beta)}{\partial \beta_0} \Big|_{(\tilde{\beta}_0, \beta_{H_0}^*)} \right\}^2 \right]:$$

• The second term in M is

$$\begin{aligned} E_{\mathbf{Y}\mathbf{X}} \left[\left\{ \frac{\partial \log f(\mathbf{Y}|\mathbf{X};\beta)}{\partial \beta_0} \Big|_{(\tilde{\beta}_0, \beta_{H_0}^*)} \right\}^2 \right] &= E_{\mathbf{Y}\mathbf{X}} \left[\left\{ \sum_{j=1}^n Y_j - \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)} \right\}^2 \right] \\ &= E_{\mathbf{Y}\mathbf{X}} \left[\left\{ \sum_{j=1}^n Y_j - np^* \right\}^2 \right] \\ &= E_{\mathbf{Y}\mathbf{X}} \left[\left(\sum_{j=1}^n Y_j \right)^2 \right] - 2np^* E_{\mathbf{Y}\mathbf{X}} \left[\left(\sum_{j=1}^n Y_j \right) \right] + n(p^*)^2 \\ &= n\text{Var}(Y_j) - 2n^2 p^* E_{\mathbf{X}}(p_j) + n(p^*)^2 \\ &= n\{p_x p_1(1 - p_1) + (1 - p_x)p_0(1 - p_0)\} - 2n^2 p^* \{p_x p_1 + (1 - p_x)p_0 - p^*\} \\ &= n\{p_x p_1(1 - p_1) + (1 - p_x)p_0(1 - p_0)\} \end{aligned}$$

LRT: Logistic Regression Example

- The first term in M is $E_{\mathbf{Y}|\mathbf{X}} \left[-\frac{\partial^2 \log f(\mathbf{Y}|\mathbf{X};\beta)}{\partial \beta_0^2} \middle|_{(\tilde{\beta}_0, \beta_{H_0}^*)} \right]^{-1} = \left\{ n \frac{\exp(\tilde{\beta}_0)}{\{1 + \exp(\tilde{\beta}_0)\}^2} \right\}^{-1}$
- Thus

$$M = \frac{p_x p_1 (1 - p_1) + (1 - p_x) p_0 (1 - p_0)}{p^* (1 - p^*)}$$

$$\text{with } p_1 = \frac{\exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\beta_{0|H_A} + \beta^*)}, p_0 = \frac{\exp(\beta_{0|H_A})}{1 + \exp(\beta_{0|H_A})} \text{ and } p^* = \frac{\exp(\tilde{\beta}_0)}{1 + \exp(\tilde{\beta}_0)}.$$

LRT: Logistic Regression Example

- As a result,

$$\begin{aligned}\lambda(\pi) &= \Delta + 1 - M \\ &= 2n \left[p_x \left\{ p_1(\beta_{0|H_A} + \beta^* - \tilde{\beta}_0) - \log \left(\frac{1 + \exp(\beta_{0|H_A} + \beta^*)}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right. \\ &\quad \left. + (1 - p_x) \left\{ p_0(\beta_{0|H_A} - \tilde{\beta}_0) - \log \left(\frac{1 + \exp(\beta_{0|H_A})}{1 + \exp(\tilde{\beta}_0)} \right) \right\} \right] \\ &\quad + 1 - \frac{p_x p_1(1 - p_1) + (1 - p_x)p_0(1 - p_0)}{p^*(1 - p^*)}\end{aligned}\tag{13}$$

- For sample size n , we have $n = \frac{\lambda(\pi) - 1 + M}{\Delta^*}$ with $\lambda(80\%) = 7.85$ and $\lambda(90\%) = 10.5$ at 5% significance level.
- For MDE, we solve (13) given p_x , $\beta_{0|H_A}$, $\tilde{\beta}_0$, $\lambda(\pi)$ and n .

Example: HPTN 037

In this example, we will use the features and estimates obtained from HPTN 037 study to calculate the following quantities based on the three hypothesis tests.

- the power of HPTN to detect an overall intervention effect corresponding to an RR of 0.5 and 0.75, given the baseline rate and sample size
- Then, the sample size needed to detect an RR of 0.5 and 0.75 with 80% and 90% power
- Then, the minimum detectable RR and risk difference (RD) given the sample size and fixed power

Example: HPTN 037 Choosing design parameters

Recall that $n = 696$ in HPTN 037. We will use estimated values from the data to guide our hypothetical exploration of study design options.

$p_x = \hat{P}(X = 1) = 336/696 = 0.483$, intended to be 0.5 by design.

Under H_A , $\hat{\beta} = (-0.022, -0.314)$.

Under H_0 , $\hat{\beta}_0 = -0.173$.

These $\hat{\beta}$ give us an estimated
 $\hat{Pr}(Y = 1|X = 0) = \hat{p}_0 = \frac{\exp\{-0.022\}}{1 + \exp\{-0.022\}} = 0.495$, the baseline outcome rate, and $\hat{Pr}(Y = 1) = \hat{p}^* = \frac{\exp\{-0.173\}}{1 + \exp\{-0.173\}} = 0.457$. This latter is the post-intervention average over $X=0$ and $X=1$.

$\hat{RR} = \exp(\hat{\beta}^*) = \exp\{-0.314\} = 0.731$

This is how we obtain approximate values for study design from existing data

Power Calculation: Wald Test

For $RR = \exp(\beta^*) = 0.5$

- We set $n = 696$, $p_x = 0.483$, $\beta_0 = -0.022$, and $\beta^* = \log(0.5)$, then $\lambda = n * \frac{(\beta^*)^2}{I_j^{-1}(\beta^*)} = 19.53$, the formula is given on slide 47.
- Power = $\Pr(X_1^2(\lambda = 19.53) > 3.84) = 99.3\%$

```
> p_x = 0.483; b0 = -0.022 ; b1=log(0.5); n=696
> # I^{-1}(\beta^*):
> Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(exp(b0)))+
+       (1/p_x)*((1+exp(b0+b1))^2/(exp(b0+b1)))
> Ib1
[1] 17.12423
> lambda_w = n*(b1^2)/Ib1
> lambda_w
[1] 19.52761
> power = 1-pchisq(3.84, 1, ncp = lambda_w, lower.tail = TRUE,)
> power
[1] 0.9930418
```

Numeric Results for Two-Sided Wald Test

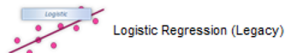
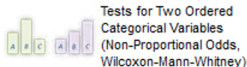
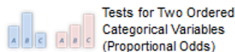
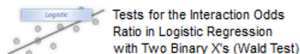
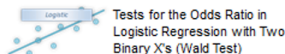
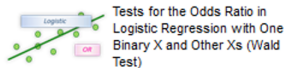
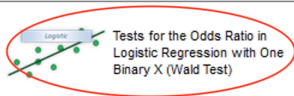
Alternative Hypothesis: $OR_{yx} \neq 1$

Power	N	Percent		P0	P1	OR _{yx}	Alpha	Beta
		X=1						
0.99303	696	48.3		0.4945	0.32846	0.5	0.05	0.00697

Wald Test-PASS results

- Using $RR = 0.5$ as an example, to obtain the PASS results, we can follow steps:
 - Step 1: Find the module: logistic regression with one binary X(Wald test)

Regression > Logistic Regression > Binary X (Wald Test) (7)



Wald Test-PASS results

- Step 2: We select it solve for Power and input n , α , p_0 , RR and p_x

Tests for the Odds Ratio in Logistic Regression with One Binary X (Wald Test) (Licensed for Student Use Only)

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Calculate

Design

Solve For: **Power**

Y = Disease 1 = Yes
X = Exposure 0 = No

Test

Alternative Hypothesis: Two-Sided

Alpha

Alpha: 0.05

Sample Size

N (Sample Size): 696

Baseline Probability

P0 [Pr(Y = 1 | X = 0)]: 0.4945

P1 or Odds Ratio

Use P1 or ORyx: ORyx

ORyx (Y,X Odds Ratio): 0.5

Prevalence of X

Percent with X = 1: 48.3

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Option Info

ORyx (Y,X Odds Ratio)

Specify one or more values of the Odds Ratio of Y and X, a measure of the effect size (event rate) that is to be detected by the study.

Odds Ratio of Y and X

This is the ratio of the odds of the outcome Y given that the exposure X = 1 to the odds of Y = 1 given X = 0.

$$\text{Odds}(Y=1|X=1) / \text{Odds}(Y=1|X=0)$$

Odds(Event A)

$$\text{Odds}(A) = \text{Prob}(A) / \text{Prob}(\text{Not } A)$$

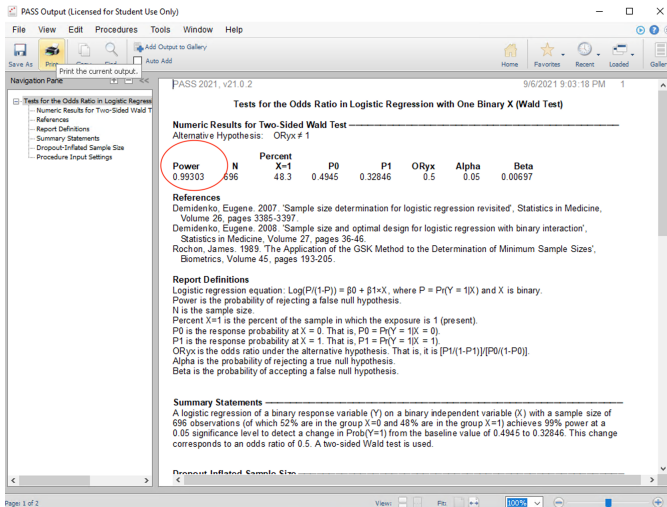
Input

You can enter a single value such as

1.5

Wald Test-PASS results

- Step 3: Click "Calculate", we will have the results



Power Calculation: Wald Test

For $RR = \exp(\beta^*) = 0.75$

- We let $n = 696$, $p_x = 0.483$, $\beta_0 = -0.022$, and $\beta^* = \log(0.75)$, then $\lambda = n * \frac{(\beta^*)^2}{I_j^{-1}(\beta^*)} = 3.551$,
- $\text{power} = Pr(X_1^2(\lambda = 3.551) > 3.84) = 47.01\%$

```
> p_x = 0.483; b0 = -0.022 ; b1=log(0.75); n=696
> # I^{-1}(\beta^*):
> Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(exp(b0))) +
+      (1/p_x)*((1+exp(b0+b1))^2/(exp(b0+b1)))
> Ib1
[1] 16.2196
> lambda_w = n*(b1^2)/Ib1
> lambda_w
[1] 3.551359
> power = 1-pchisq(3.84, 1, ncp = lambda_w, lower.tail = TRUE,)
> power
[1] 0.4701334
```

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis: $OR_{yx} \neq 1$

Power	N	Percent X=1	P0	P1	OR _{yx}	Alpha	Beta
0.46999	696	48.3	0.4945	0.42319	0.75	0.05	0.53001

Power Calculation for the Score Test

For $RR = \exp(\beta^*) = 0.5$, then $\beta^* = \log(0.5)$

- We let $n = 696$, $p_x = 0.483$, $\tilde{\beta}_0 = -0.173$, $\beta_{0|H_A} = -0.022$, then
$$\lambda = \xi^2 / I^*(\beta^*) = n \times \frac{p_x(1-p_x)(p_1-p_0)^2}{p_x p_0(1-p_0) + (1-p_x)p_1(1-p_1)} = 20.41,$$
- power = $Pr(X_1^2(\lambda = 20.41) > 3.84) = 99.5\%$

```
> p_x = 0.483; b00 = -0.173; b0a = -0.022; b1=log(0.5) ; n=696#
> p_0 =exp(b0a)/(1+exp(b0a)); # p_y0 =exp(b00)/(1+exp(b00));
> p_1 = exp(b0a+b1)/(1+exp(b0a+b1))
> lambda_s = n*(p_x*(1-p_x)*(p_1-p_0)^2)/(p_x*p_0*(1-p_0)+ (1-p_x)*p_1*(1-p_1))
> lambda_s
[1] 20.40859
> power = 1-pchisq(3.84, 1, ncp = lambda_s, lower.tail = TRUE, log.p = FALSE)
> power
[1] 0.9947361
> |
```

For $RR = \exp(\beta^*) = 0.75$, then $\beta^* = \log(0.75)$

- We let $n = 696$, $p_x = 0.483$, $\tilde{\beta}_0 = -0.173$, $\beta_{0|H_A} = -0.022$, then
$$\lambda = \xi^2 / I^*(\beta^*) = n \times \frac{p_x(1-p_x)(p_1-p_0)^2}{p_x p_0(1-p_0) + (1-p_x)p_1(1-p_1)} = 3.579,$$
- power = $Pr(X_1^2(\lambda = 3.579) > 3.84) = 47.3\%$

Power Calculation: LRT

For $RR = \exp(\beta^*) = 0.5$

- We let $n = 696$, $p_x = 0.483$, $\beta_{0|H_A} = -0.022$, $\tilde{\beta}_0 = -0.346$ and $\beta^* = \log(0.5)$, $\Delta^* = 0.0286$ and $M = 0.972$ with the formula of Δ^* on slide 62 and the formula of M on slide 64.
- $\lambda = 19.908$ using the formula on slide 65, and power = 99.38%

```
> p_x = 0.483; b0a = -0.022; b1=log(0.5); n=696
> p0 =exp(b0a)/(1+exp(b0a)); p1 = exp(b0a+b1)/(1+exp(b0a+b1))
> p_star = (1-p_x)*p0+p_x*p1; b_tilde = log(p_star/(1-p_star))
> b_tilde
[1] -0.3462009
> D_star = 2*( p_x*(p1*(b0a+b1)-p_star*b_tilde -
+ log((1+exp(b0a+b1))/(1+exp(b_tilde))) ) )
+ (1-p_x)* (p0*b0a-p_star*b_tilde-
+ log((1+exp(b0a))/(1+exp(b_tilde))) ) )
> D_star
[1] 0.0285624
> M_t = (p_x*(1-p1)*p1 +(1-p_x)*p0*(1-p0))/
+ (p_star *(1-p_star) )
> M_t
[1] 0.97163
> lambda = n*D_star+1-M_t
> lambda
[1] 19.9078
> power=1-pchisq(3.84, 1, ncp = lambda,lower.tail = TRUE)
> power
[1] 0.9938292
```

For $RR = \exp(\beta^*) = 0.75$

- We let $n = 696$, $p_x = 0.483$, $\beta_{0|H_A} = -0.022$, $\tilde{\beta}_0 = -0.160$ and $\beta^* = \log(0.75)$, $\Delta^* = 0.005$ and $M = 0.995$
- $\lambda = 3.567$ and power = 47.18%

Sample Size Calculation: For Wald Test

When $RR = \exp(\beta^*) = 0.50$

- We let power = 80%, $p_x = 0.483$, $\beta_0 = -0.022$, and $\beta^* = \log(0.5)$, then $n = 280$ using the formula on page 47.
- We let power = 90%, then $n = 375$.

```
p_x = 0.483;b0=-0.022;b1 = log(0.5)
Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(exp(b0)))+
      (1/p_x)*((1+exp(b0+b1))^2/(exp(b0+b1)))
n_80=7.85/((b1^2)/Ib1)
n_90 = 10.5/((b1^2)/Ib1)
n_80
[1] 279.7885
n_90
[1] 374.2394
```

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis: ORyx ≠ 1

Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.80035	280	48.3	0.4945	0.32846	0.5	0.05	0.19965

Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.90038	375	48.3	0.4945	0.32846	0.5	0.05	0.09962

Wald Test - PASS Results

- We use same module: logistic regression with one binary X (Wald test), but in Step 2, we select to solve sample size. Here we input power, α , p_0 , OR and p_x

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Calculate

Design

Solve For: Sample Size

Y = Disease 1 = Yes
X = Exposure 0 = No

Test

Alternative Hypothesis: Two-Sided

Power and Alpha

Power: 0.8

Alpha: 0.05

Baseline Probability

P0 [Pr(Y = 1 | X = 0)]: 0.4945

P1 or Odds Ratio

Use P1 or ORyx: ORyx

ORyx (Y,X Odds Ratio): 0.5

Prevalence of X

Percent with X = 1: 48.3

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Option Info

Power

Power is the probability of rejecting the null hypothesis when it is false. Power is equal to 1 - Beta, so specifying power implicitly specifies beta.

Beta is the probability of obtaining a false negative with the statistical test. That is, it is the probability of accepting a false null hypothesis.

Range

The valid range is 0 to 1. Usually, a power between 0.8 and 0.95 is desired.

Recommended

Different disciplines have different standards for setting power. The most common choice is 0.90, but 0.80 is also popular.

Wald Test - PASS Results

- After click "Calculate", we can obtain the results:

The screenshot displays the PASS 2021, v21.0.2 interface. The main window shows the 'Tests for the Odds Ratio in Logistic Regression with One Binary X (Wald Test)' results. A red circle highlights the 'Power' value of 0.80036 in the 'Numeric Results for Two-Sided Wald Test' table. The table includes columns for Power, N, Percent X=1, P0, P1, ORyx, Alpha, and Beta. Below the table, the 'References' section lists three sources: Demidenko (2007), Demidenko (2008), and Rochon (1989). The 'Report Definitions' section provides mathematical definitions for the variables used in the test. The 'Summary Statements' section concludes that a logistic regression model with a sample size of 280 observations achieves 80% power at a 0.05 significance level to detect a change in the probability of Y=1 from 0.4945 to 0.32846.

Tests for the Odds Ratio in Logistic Regression with One Binary X (Wald Test)

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis: $OR_{yx} \neq 1$

Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.80036	280	48.3	0.4945	0.32846	0.5	0.05	0.19965

References

Demidenko, Eugene. 2007. 'Sample size determination for logistic regression revisited', *Statistics in Medicine*, Volume 26, pages 3385-3397.

Demidenko, Eugene. 2008. 'Sample size and optimal design for logistic regression with binary interaction', *Statistics in Medicine*, Volume 27, pages 36-46.

Rochon, James. 1989. 'The Application of the GSK Method to the Determination of Minimum Sample Sizes', *Biometrics*, Volume 45, pages 193-205.

Report Definitions

Logistic regression equation: $\text{Log}[P/(1-P)] = \beta_0 + \beta_1 \times X$, where $P = \text{Pr}(Y = 1|X)$ and X is binary.

Power is the probability of rejecting a false null hypothesis.

N is the sample size.

Percent $X=1$ is the percent of the sample in which the exposure is 1 (present).

P_0 is the response probability at $X = 0$. That is, $P_0 = \text{Pr}(Y = 1|X = 0)$.

P_1 is the response probability at $X = 1$. That is, $P_1 = \text{Pr}(Y = 1|X = 1)$.

OR_{yx} is the odds ratio under the alternative hypothesis. That is, it is $[P_1/(1-P_1)]/[P_0/(1-P_0)]$.

Alpha is the probability of rejecting a true null hypothesis.

Beta is the probability of accepting a false null hypothesis.

Summary Statements

A logistic regression of a binary response variable (Y) on a binary independent variable (X) with a sample size of 280 observations (of which 52% are in the group $X=0$ and 48% are in the group $X=1$) achieves 80% power at a 0.05 significance level to detect a change in $\text{Prob}(Y=1)$ from the baseline value of 0.4945 to 0.32846. This change corresponds to an odds ratio of 0.5. A two-sided Wald test is used.

For Wald Test

When $RR = \exp(\beta^*) = 0.75$,

- We let power = 80%, $p_x = 0.483$, $\beta_0 = -0.022$, and $\beta^* = \log(0.75)$, then $n = 1539$
- We let power = 90%, then $n = 2058$

```
> p_x = 0.483;b0 = -0.022;b1 = log(0.75)
> Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(exp(b0)))+
+      (1/p_x)*((1+exp(b0+b1))^2/(exp(b0+b1)))
> n_80=7.85/((b1^2)/Ib1)
> n_90 = 10.5/((b1^2)/Ib1)
> n_80
[1] 1538.453
> n_90
[1] 2057.804
```

Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.8002	1539	48.3	0.4945	0.42319	0.75	0.05	0.1998
Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.9001	2060	48.3	0.4945	0.42319	0.75	0.05	0.0999

Sample Size Calculation: For Score Test

When $RR = \exp(\beta^*) = 0.50$

- We let power = 80%, $p_x = 0.483$, $\beta_{0|H_0} = -0.173$, $\beta_{0|H_A} = -0.022$, and $\beta^* = \log(0.5)$, then $n = 268$ using formula on page 56 given p_1 (p_y1 in the R code) and p_0 (p_y0 in the R code).

```
> p_x = 0.483;b00 = -0.173;b0a = -0.022;b1=log(0.5) #  
> p_y0 = exp(b0a)/(1+exp(b0a))  
> p_y1 = exp(b0a+b1)/(1+exp(b0a+b1))  
> n_80=7.85/((p_x*(1-p_x)*(p_y1-p_y0)^2)/  
+ (p_x*p_y0*(1-p_y0)+ (1-p_x)*p_y1*(1-p_y1)))  
>  
> n_90= 10.5/((p_x*(1-p_x)*(p_y1-p_y0)^2)/  
+ (p_x*p_y0*(1-p_y0)+ (1-p_x)*p_y1*(1-p_y1)))  
> n_80  
[1] 267.7108  
> n_90  
[1] 358.0845
```

- We let power = 90%, then $n = 359$

When $RR = \exp(\beta^*) = 0.75$,

- We let power = 80%, $p_x = 0.483$, $\beta_{0|H_0} = -0.173$, $\beta_{0|H_A} = -0.022$, $\beta^* = \log(0.75)$, then $n = 1527$
- We let power = 90%, then $n = 2043$

Sample Size Calculation: For LRT

When $RR = \exp(\beta^*) = 0.50$

- We let power = 80%, $p_x = 0.483$, $\beta_{0|H_A} = -0.022$, $\tilde{\beta}_0 = -0.3462$ and $\beta^* = \log(0.5)$, then $n = 274$. We let power = 90%, then $n = 367$ using the formula for n on slide 65 with the formula for Δ^* on slide 62 and the formula for M on page 64.

```
> p_x = 0.483; b0a = -0.022; b1=log(0.5); n=696
> p0 =exp(b0a)/(1+exp(b0a)); p1 = exp(b0a+b1)/(1+exp(b0a+b1))
> p_star = (1-p_x)*p0+p_x*p1; b_tilde = log(p_star / (1-p_star ))
> D_star = 2*( p_x*(p1*(b0a+b1)-p_star*b_tilde - log( (1+exp(b0a+b1))/(1+exp(b_tilde)) ) )
+ (1-p_x)* (p0*b0a-p_star*b_tilde- log((1+exp(b0a))/(1+exp(b_tilde)) ) ) )
> D_star
[1] 0.0285624
> M_t = (p_x*(1-p1)*p1 +(1-p_x)*p0*(1-p0))/(p_star *(1-p_star ) )
> M_t
[1] 0.97163
> #power = 80%
> n_80=(7.85-1+M_t)/D_star
> n_80
[1] 273.8436
> #power = 90%
> n_90=(10.5-1+M_t)/D_star
> n_90
[1] 366.6229
```

When $RR = \exp(\beta^*) = 0.75$,

- We let power = 80%, then $n = 1533$. We let power = 90%, then $n = 2051$.

MDE Calculation: Wald Test

- We let power = 80%, $n = 696$, $p_x = 0.483$, $\beta_0 = -0.022$, then MDE $\beta^* = -0.431$ then $RR = \exp(\beta^*) = 0.650$

```
p_x = 0.483; b0 = -0.022; n=696
power.eq <-function(b1,para){
  p_x = para[1]
  b0=para[2]
  n=para[3]
  Ib1 = (1/(1-p_x))*((1+exp(b0))^2/(1+exp(b0))) +
  (1/p_x)*((1+exp(b0+b1))^2/(1+exp(b0+b1)))
  lambda_weq = n*(b1^2)/Ib1-7.85
}
find para<-multroot(power.eq , start =-0.1, para = c(0.483,-0.022,696), positive = FALSE)
```

- We let power = 90%, MDE $\beta^* = -0.499$, then $RR = \exp(\beta^*) = 0.607$
- Using PASS module: logistic regression with one binary X(Wald test), we can obtain MDE by fixing n , and trying different values of RR to find the one that gives power of 80% or 90% . For example:
 - We first let $n = 696$, $p_x = 0.483$, and try $RR = 0.7$, then the power = 64.4%

Numeric Results for Two-Sided Wald Test
Alternative Hypothesis: $OR_{yx} \neq 1$

Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.64406	696	48.3	0.4945	0.40645	0.7	0.05	0.35594

MDE Calculation: Wald Test

- Then we let try $RR = 0.650$, then the power = 80%

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis: $OR_{yx} \neq 1$

Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.80011	696	48.3	0.4945	0.3887	0.65	0.05	0.19989

- When we let $RR = 0.607$, then the power = 89.85%

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis: $OR_{yx} \neq 1$

Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.89851	696	48.3	0.4945	0.37257	0.607	0.05	0.10149

- When we let $RR = 0.606$, then the power = 90.03%

Numeric Results for Two-Sided Wald Test

Alternative Hypothesis: $OR_{yx} \neq 1$

Power	N	Percent X=1	P0	P1	ORyx	Alpha	Beta
0.90033	696	48.3	0.4945	0.37218	0.606	0.05	0.09967

- Base on our attempts, we can conclude that, when we let $RR = 0.650$, the power=80%; when we let RR between 0.606 and 0.607, the power =90%.

MDE Calculation: Score Test

- We let power = 80%, $n = 696$, $p_x = 0.483$, $\beta_0 = -0.022$, then MDE $\beta^* = -0.427$ and $RR = \exp(\beta^*) = 0.652$. The formula for calculating MDE is in page 56. p_{y1} and p_{y0} in the code are p_1 and p_0 , respectively.

```
library(rootSolve)
p_x = 0.483; b00=-0.173; b0a = -0.022; n=696;

power.eq1 <-function(b1,para){
  b00 = para[1]; b0a=para[2]; n=para[3]; p_x= para[4]
  p_y0 = exp(b0a)/(1+exp(b0a))
  p_y1 = exp(b0a+b1)/(1+exp(b0a+b1))

  lambda_0= n*(p_x*(1-p_x)*(p_y1-p_y0)^2)/
    (p_x*p_y0*(1-p_y0)+ (1-p_x)*p_y1*(1-p_y1))-7.85

  # power = power
}
find.para<-multiroot(power.eq1, start =-0.1,
  para = c(-0.173,-0.022,696,0.483), positive = FALSE)
```

- We let power = 90%, MDE $\beta^* = -0.495$, then $RR = \exp(\beta^*) = 0.610$

MDE Calculation: LRT Test

- power = 80%, $p_x = 0.483$, $\beta_{0|H_A} = -0.022$, $\tilde{\beta}_0 = -0.3462$, $n = 696$, then $\beta^* = -0.429$, $RR = \exp(\beta^*) = 0.651$, the formula for calculating MDE is in page 65.
- We let power = 90%, MDE $\beta^* = -0.498$, then $RR = \exp(\beta^*) = 0.608$

```
library(rootSolve)
p_x = 0.483; b0a = -0.022; n=696
power.eql <-function(b1,para){
  b0a=para[1]; n=para[2]; p_x= para[3]
  p0 =exp(b0a)/(1+exp(b0a)); p1 = exp(b0a+b1)/(1+exp(b0a+b1))
  p_star = (1-p_x)*p0+p_x*p1; b_tilde = log(p_star / (1-p_star ))
  D_star = 2*( p_x*(p1*(b0a+b1)-p_star*b_tilde -log( (1+exp(b0a+b1))/(1+exp(b_tilde)) ) )
    +(1-p_x)* (p0*b0a-p_star*b_tilde- log((1+exp(b0a))/(1+exp(b_tilde)) ) ) )
  M_t = (p_x*(1-p1)*p1 +(1-p_x)*p0*(1-p0))/(p_star *(1-p_star ) )
  lambda_leq = n*D_star+1-M_t -7.85
}
find para<-multiroot(power.eql, start =-0.1,
  para = c(b0a,n,p_x), positive = FALSE)
find para$root
[1] -0.4289922
exp(-0.4289922)
[1] 0.651165
```

Compare three tests

Table 2: Power Analysis Results of HPTN 037 data

		Wald Test	Score Test	LRT
Power	RR=0.5	99.30%	99.47%	99.38%
	RR=0.75	47.01%	47.30%	47.18%
Sample Size	RR=0.5			
	Power =80%	280	268	274
	Power = 90%	375	359	367
	RR=0.75			
	Power =80%	1539	1527	1533
	Power = 90%	2058	2043	2051
MDE	Power =80%	0.650	0.652	0.651
	Power = 90%	0.607	0.610	0.608

Notation

i : i th group, $i=1,2$

j : j th observation

β^* : the parameter of interest

β_0 : nuisance parameter

$\beta = (\beta_0, \beta^*)^T$: the vector of parameters

q : number of parameter in β

$\beta_{H_0} = (\beta_{0|H_0}, \beta_{0|H_0}^*)^T$: the vector of parameters under H_0

$\beta_{H_A} = (\beta_{0|H_A}, \beta_{0|H_A}^*)^T$: the vector of parameters under H_A

μ_1 and μ_2 : two sample means

n : total sample size

$\mathbf{Y} = (Y_1, \dots, Y_n)^T$: the outcomes for participants $j = 1, \dots, n$

$\mathbf{X} = (X_1, \dots, X_n)^T$: the intervention/ treatment status for participants

$Z^2(\mathbf{Y}, \mathbf{X}; \beta)$: test statistic which is a function of the data $\{\mathbf{Y}, \mathbf{X}\}$.

λ : non-central parameter of χ^2 distribution

χ_1^2 : central χ^2 distribution with $DF = 1$

$\chi_1^2(\lambda)$: non-central χ^2 distribution with $DF = 1$ and λ

$p_j = \frac{\exp(\beta_0 + \beta^* X_1)}{1 + \exp(\beta_0 + \beta^* X_1)}$: the probability of $P(Y_j = 1 | X_j; \beta)$ under logistic regression model

Notation

p_0 : $Pr(Y = 1|X = 0)$ under H_A

p_1 : $Pr(Y = 1|X = 1)$ under H_A

p^* : $Pr(Y = 1)$ under H_A

U_j : first derivative of the log-likelihood for single observation

p_x : given $X \sim \text{Bern}(p_x)$, $p_x = Pr(X = 1)$

$\mathbf{U}$: $\sum_j^n U_j(Y_j, X_j; \beta)$

$\mathbf{I}$: $\mathbf{I} = \text{Var}(\mathbf{U})$, also be the negative expectation of information matrix of all observations

$I_j(\beta)$: the negative expectation of information matrix for a single observation

$I_j^{-1}(\beta^*)$: the $r - r$ th element of the covariance matrix of $\text{Var}(\beta)$ based on the $I_j(\beta)$.

π : the statistical power of a test

α : the significant level of a test

U_{β^*} : the r th element in the score function $\mathbf{U}$

U_{β_0} : the element in the score function for β_0

Notation

$\hat{V}_{r-r}(\mathbf{Y}, \mathbf{X})$: the estimated of the variance of $U_{\beta^*}(\hat{\beta}_{0|H_0}, \beta^*)$

ξ : the expectation of the approximation of $U_{\beta^*}(\hat{\beta}_{0|H_0}, \beta^*)$ with respect to $(\mathbf{Y}, \mathbf{X})$ evaluated as $\beta_{H_0}^*$ and $\tilde{\beta}_0$.

$I^*(\beta^*)$: the variance of the approximation of $U_{\beta^*}(\hat{\beta}_{0|H_0}, \beta^*)$ with respect to $(\mathbf{Y}, \mathbf{X})$ evaluated as $\beta_{H_0}^*$ and $\tilde{\beta}_0$.

M : a matrix which trace is an approximation of the expected value of the second term in (11).

Δ : the expected value of the third term in (11)

Z_{λ}^2 : a random variable following $\chi_1^2(\lambda)$ distribution with DF=1, and non-centrality parameter λ .